UG

UrbanGaon®

INTELLIGENT VOICE & CALENDAR TODO PLATFORM

ENGINEERING & EXECUTIVE DOCUMENTATION

API Testing Manual & In-App Email Synchronization Guide

Comprehensive Postman testing protocol covering authentication, deliverable lifecycles, and direct synchronization with the website's live SMTP email dispatcher and audit ledger.

SYSTEM VERSION **v2.4.0 (Enterprise Release)**	DATE & ENVIRONMENT **November 2026 • Localhost / Intranet**
PRIMARY EXECUTIVE **Mr. Sukh Sagar (Founder)**	LIVE DISPATCHER EMAIL **techdesk@urbangaon.com**
SMTP GATEWAY **smtp.gmail.com:465 (Direct SSL)**	COLLECTION FILE **postman_collection.json (v2.1)**

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1. System Architecture & Email Synchronization Map

The UrbanGaon platform integrates an automated, transactional email notification engine (Flowchart 20) with live MongoDB Atlas data operations. Every critical API mutation dispatches verified emails and records an entry in both the **Immutable Audit Log** and **Email Dispatch Ledger**.

SMTP Dispatcher Configuration (Live from .env)

Configuration Key	Value in System	Operational Role
`SMTP_HOST`	`smtp.gmail.com`	High-reliability Google SMTP server
`SMTP_PORT`	465 (SSL Secured)	Direct encrypted transmission
`SMTP_USER`	`techdesk@urbangaon.com`	Corporate executive dispatcher account
`SMTP_FROM`	`"UrbanGaon Executive Dispatcher" <techdesk@urbangaon.com>`	Authenticated sender identifier
`NEXT_PUBLIC_APP_URL`	`http://192.168.1.44:3000`	Direct hyperlink embedded in all outgoing emails

Live Synchronization Event Matrix

Triggering API Route	Event Type	Recipient in System	Dispatched Email Subject	In-App Ledger Log
`POST /api/auth/login`	`2FA_OTP`	`techdesk@urbangaon.com` / `ceo@urbangaon.com`	Your UrbanGaon Two-Step Verification Code: [OTP]	Logged in DB & console
`POST /api/tasks`	`TASK_ASSIGNED`	Assignee (e.g. `ravindra@urbangaon.com`)	New Deliverable Assigned: "[Title]"	`EmailLog` + `AuditLog`
`PATCH /api/tasks/:id` (Submit)	`TASK_REVIEW`	Founder (`ceo@urbangaon.com`)	Deliverable Pending Final Approval: "[Title]"	`InAppNotification` + Banner
`PATCH /api/tasks/:id` (Approve)	`STATUS_CHANGED`	All Task Assignees	Deliverable Approved by Mr. Sukh Sagar: "[Title]"	`EmailLog` + Audit Trail
`POST /api/tasks/evaluate-overdue`	`OVERDUE_ALERT`	Assignees & Department Heads	Urgent Action Required: Deliverable Overdue	Daily Overdue Ledger

Triggering API Route	Event Type	Recipient in System	Dispatched Email Subject	In-App Ledger Log
`GET /api/cron?action=digest`	`DAILY_DIGEST`	All Active Corporate Staff	UrbanGaon Morning Deliverables Digest — [Date]	Digest Batch Summary

2. Step-by-Step API Testing Protocols

`POST /api/auth/login`

Test 1: Executive 2FA Authentication (Synced with OTP Email)

Executive accounts require a two-step verification cycle. When Step 1 credentials pass, the server dispatches a real-time 6-digit OTP to `techdesk@urbangaon.com`.

Step 1: Send Credentials

```
POST {{baseUrl}}/api/auth/login
Content-Type: application/json

{
  "email": "ceo@urbangaon.com",
  "password": "admin"
}
```

Expected Step 1 Response (2FA Prompt):

```
{
  "success": false,
  "requires2FA": true,
  "role": "SUPER_ADMIN",
  "message": "Two-step verification required. A 6-digit verification code has been dispatched in real t
}
```

Step 2: Submit with 2FA Code

Supply the 6-digit OTP received in email, or use the Founder development bypass code `123456`:

```
POST {{baseUrl}}/api/auth/login
Content-Type: application/json

{
  "email": "ceo@urbangaon.com",
  "password": "admin",
  "twoFactorCode": "123456"
}
```

Success Response: Returns JWT Bearer token and sets secure HTTP-only cookie `urbangaon_auth_token`. Copy this token into your Postman collection variable `{{authToken}}`.

`POST /api/tasks`

Test 2: Create Deliverable (Synced with Task Assignment Email)

Creates an official task and dispatches an immediate assignment email via Gmail SMTP to all designated assignees.

```
POST {{baseUrl}}/api/tasks
Content-Type: application/json
Authorization: Bearer {{authToken}}

{
  "title": "Vendor payment verification and ledger reconciliation",
  "scheduledDate": "2026-11-06",
  "time": "05:00 PM",
  "priority": "HIGH",
  "department": "Procurement & Materials",
  "project": "Jaipur Expansion Project",
  "location": "Jaipur Corporate Office",
  "assignees": [
    {
      "id": "mem-2",
      "name": "Ravindra Singh",
      "email": "ravindra@urbangaon.com",
      "department": "Procurement & Materials",
      "status": "Not Started"
    }
  ],
  "creator": {
    "id": "mem-ops-admin",
    "name": "Operations Admin",
    "email": "ops@urbangaon.com",
    "role": "ADMIN"
  }
}
```

✓ Verification: Inspect `GET /api/notifications/email-logs` to see the recorded dispatch with subject *"New Deliverable Assigned: Vendor payment verification..."*.

3. Governance, Approval & Status Sync Testing

`PATCH /api/tasks/:id`

Test 3: Employee Submits Deliverable for Founder Sign-Off

Assignee marks work completed, transitioning deliverable to `Review` status with badge **AWAITING MR SUKH SAGAR APPROVAL** and triggering an executive alert.

```
PATCH {{baseUrl}}/api/tasks/task-1
Content-Type: application/json
Authorization: Bearer {{authToken}}

{
  "status": "Review",
  "progressPercent": 95,
  "pendingCeoApproval": true,
  "ceoApprovalRequestedBy": "Ravindra Singh",
  "remarks": "Invoices checked and vendor balance reconciled. Ready for Founder sign-off."
}
```

> **Website Display Sync:** Deliverable instantly displays the orange pill `AWAITING MR SUKH SAGAR APPROVAL` on both Company Dashboard and Founder View.

`PATCH /api/tasks/:id`

Test 4: Founder Grants Final Approval (Completion Sync)

Executive signs off on the deliverable. Transitions status to `Done` and dispatches completion emails to all assignees.

```
PATCH {{baseUrl}}/api/tasks/task-1
Content-Type: application/json
Authorization: Bearer {{authToken}}

{
  "status": "Done",
  "progressPercent": 100,
  "pendingCeoApproval": false,
  "ceoApprovedBy": "Mr. Sukh Sagar",
  "ceoApprovedAt": "2026-11-04T12:00:00.000Z",
  "remarks": "Final approval granted by Mr. Sukh Sagar. Marked Done."
}
```

✓ Email Dispatched: *"Mr. Sukh Sagar has granted Final Approval. Your deliverable is officially Completed (Done)!"*

`GET /api/cron?action=digest`

Test 5: Trigger Morning Deliverables Email Digest (08:30 AM IST)

Executes the morning digest engine, compiling active tasks and sending individual email summaries to all employees.

```
GET {{baseUrl}}/api/cron?action=digest
```

```
{
  "success": true,
  "action": "digest",
  "sentCount": 18,
  "message": "Morning digest executed: 18 emails dispatched."
}
```

4. Postman Environment Setup & UI Verification

Importing the Preconfigured Collection

1. Open Postman and click **Import** in the top-left corner.
2. Choose the file located at: `d:\TODO Dashboard New updated Version\postman_collection.json`.
3. The collection **"UrbanGaon AI Todo Platform API"** will appear in your sidebar with 8 pre-built folders.

Collection Variables Reference

Variable Name	Default Value	Description
`baseUrl`	`http://localhost:3000`	Server target (switch to `http://192.168.1.44:3000` for network testing)
`authToken`	*Auto-populated after login*	JWT token attached in `Authorization: Bearer {{authToken}}`
`testTaskId`	`task-1`	Target deliverable ID for update/delete requests

How to Verify Live Email Dispatch Inside Website UI

1. Postman Trigger
Send API Request (e.g. create task or grant approval)
→
2. Gmail SMTP Server
Real email delivered to corporate inbox
→
3. Website Email Log
Verify record at `/api/notifications/email-logs`
→
4. Immutable Audit Log
Verify timestamp & actor in website Audit view

Verifying Directly via API

To inspect all emails sent by the system without opening Gmail, query the email log endpoint:

`GET /api/notifications/email-logs`

```
{
  "success": true,
  "count": 42,
  "data": [
    {
      "to": "ravindra@urbangaon.com",
      "subject": "New Deliverable Assigned: Vendor payment verification",
      "eventType": "TASK_ASSIGNED",
      "status": "SENT",
      "timestamp": "2026-11-04T12:05:00.000Z"
    }
  ]
}
```

Validated against ISO/IEC 27001 data integrity standards & executive governance flowcharts.